

Formalism of the Scalar-Angular-Torsion (SAT) Framework: 4D Filament Dynamics and the Zero-Parameter Economy

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We present the formal mathematical structure of the Scalar-Angular-Torsion (SAT) framework, which replaces point-particle mechanics with a discretized 4D hypersphere lattice (the 24-cell HSUCV). In this ontology, matter is defined as persistent 4D superhelical filaments, and mass emerges as projective resistance against universal expansion. We derive the master Lagrangian governing filament dynamics, the universal mass scaling laws for baryons and leptons, and the geometric thresholds for relativistic obscuration. Technical data for the Helium-3 holotype and nuclear topology mapping for Li-4, Si-26, Ge, and Tc are provided as evidence of mathematical closure and the zero-parameter economy.

I. FUNDAMENTAL ANCHORS AND CONSTANTS

The SAT framework is anchored by a single lattice scale, l_f , from which all physical observables derive via the geometry of the 24-cell HSUCV lattice. Table I summarizes the foundational constants used in the zero-parameter derivations.

TABLE I. Fundamental SAT Constants derived from the HSUCV Lattice.

Parameter	Symbol	Value (Units)
Filament Scale	l_f	$\approx 0.7937 \text{ fm}$
Universal Mass Anchor	m_0	$\approx 1.0073 \times 10^{-27} \text{ kg}$
Projection Constant	B	$\approx 0.2387 \text{ rad}$
Obscuration Constant	θ_{obs}	$\approx 0.246 \text{ rad (14.1}^\circ\text{)}$
Topological Mode Density	ρ_{embed}	$\approx 10^{-19}$
Electromagnetic Residual	k	0.25 (geometric)
Lepton Gear Suppression	B_4	$B/4$

II. MATHEMATICAL FORMALISM

A. Universal Indicatrix (UI) Mapping

The UI maps 4D filament trajectories x_0^ν onto observable 3D projections y^μ through a non-linear geometric derivation involving scale $r(\lambda)$ and the rotation tensor $R_\nu^\mu(\lambda)$:

$$y^\mu(\lambda) = r(\lambda) R_\nu^\mu(\lambda) x_0^\nu \quad (1)$$

The vacuum baseline is defined by a ground-state geometric effort of exactly 0.5 units:

$$L_{UI} = \frac{1}{2} \dot{r}^2 + \frac{1}{2} r^2 \Omega_{\mu\nu} \Omega^{\mu\nu} = 0.5 \quad (2)$$

B. The Master SAT Lagrangian

The dynamics of n th-order superhelical filaments are governed by the Master Lagrangian, which integrates kinetic energy, inter-filament braiding, and internal torsion:

$$L_{SAT} = \sum_i \sum_o \left[\frac{1}{2} \mu_i \left| \frac{d\lambda}{dH} \right|^2 + \sum_j \kappa F_{\text{braid}}(H_i, H_j) + \alpha \left(\frac{d\lambda}{dH} \cdot T \right)^2 + V_{geom}(\lambda) \right], \quad (3)$$

where $V_{geom}(\lambda)$ accounts for global curvature and the Braid-Smoothing Metric:

$$V_{geom}(\lambda) = -\frac{k}{R \sin(\chi)} + \text{Metric}_{\text{smoothing}} \quad (4)$$

III. PARTICLE TOPOLOGY AND MASS SCALING

A. Topological Charge (Q)

Mass emerges from Projective Resistance (R). The topological charge Q is defined as the sum of linking and winding numbers per HSUCV node. Stability is governed by the UV Finiteness Lock, where persistent intersections are capped at $Q \leq 3$.

B. Scaling Laws

Baryon mass ($Q \geq 3$) incorporates non-linear smoothing corrections:

$$m_B = 2B^Q \cdot m_0 - \text{Metric}_{\text{smoothing}} \quad (5)$$

Lepton mass ($Q = 1$) utilizes the fractional gear suppression:

$$m_L = Q \cdot m_0 \cdot B_4 \quad (6)$$

IV. RELATIVISTIC AND ATOMIC PHENOMENA

A. Critical Velocity and Obscuration

At the critical velocity v_{crit} , matter precesses out of the 3D slice, leading to the Dark Matter Illusion:

$$v_{crit} = B \cdot c \approx 0.2387c \quad (7)$$

Transitions across the θ_{obs} threshold manifest as discrete, achromatic phase snaps of exactly 0.246 rad.

B. Atomic Energy Levels

The hydrogenic spectrum emerges from the S^3 geometry, including the $1/R^2$ cosmic correction:

$$E_n = -\frac{2\hbar^2\mu k^2}{n^2} + O\left(\frac{1}{R^2}\right) \quad (8)$$

V. TECHNICAL NUCLEAR DATA

The Helium-3 nucleus serves as the framework's holotype due to its combinatorial bareness. Table II maps the topological charge and projected mass for several key isotopes.

TABLE II. SAT Nuclear Topology and Mass Scaling.

Isotope	Class	Charge (Q)	Mass (kg)
He-3 (Anchor)	Holotype	3.0	3.3577×10^{-28}
Li-4	Deviation	4.0	2.5183×10^{-28}
Si-26	Spectrum Block	26.0	3.8742×10^{-29}
Germanium	Dense Weave	72.6	1.3875×10^{-29}
Technetium	Torsion Anomaly	98.0	1.0279×10^{-29}

A. The He-3 Magnetic Moment

The He-3 magnetic moment is derived as a geometric dual of the 4D current $J_{\mu\nu\rho}$ weaving through the HSUCV vertex, locked by a 270° quarter-turn holonomy:

$$\mu_{He3} \approx -2.127\mu_N \quad (9)$$

VI. CONCLUSION

The SAT framework provides a mathematically closed system that derives emergent 3D physics from 4D mechanical rotations. Falsification targets include the detection of 0.246 rad phase shifts in pulsar timing and the verification of the Jarlskog Residual ($J_{eff} \approx -3.3 \times 10^{-2}$) at the Helium-3 nuclear vertex.

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